Nebojsa Jovanovic

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Professional Summary

Mechanical Engineer with a Bachelor of Applied Science in Mechanical Engineering and experience at Hatch supporting multidisciplinary engineering projects, package engineering, industrial systems, commissioning, and technical project execution. Background includes industrial infrastructure, mineral processing, energy systems, tunnel fire ventilation design, systems integration, and engineering workflow automation. Experienced in coordinating technical deliverables, working with engineering and construction stakeholders, organizing project documentation, and applying analytical problem-solving to complex engineering challenges. Combines mechanical engineering fundamentals with software development and applied AI capabilities to improve technical workflows, automate repetitive tasks, and support engineering productivity. Hatch Engineering Entrance Scholarship recipient with experience contributing to large-scale industrial projects and collaborating across engineering, construction, commissioning, and technology teams.

Technical Skills

- **Mechanical Engineering:** Mechanical Engineering Principles, Systems Design, Package Engineering, Tunnel Fire Ventilation, Industrial Systems, Equipment and System Integration
- **Engineering Project Delivery:** Multidisciplinary Coordination, Technical Documentation, Systems Handover, Commissioning Support, Deliverable Tracking, Technical Problem-Solving, Project Execution
- **Design and Technical Coordination:** Engineering Requirements Interpretation, Systems Integration, Technical Information Management, Design Coordination Support, Engineering Documentation Review
- **Industrial and Infrastructure Projects:** Mineral Processing, Lithium and Copper Processing, Potash Processing, Energy Systems, Tunnel Infrastructure, Industrial Automation
- **Engineering Software and Documentation:** SmartPlant Foundation, Smart Completions, SharePoint, Azure DevOps
- **Software and Automation:** Python, C++, C#, Go, JavaScript, TypeScript, SQL, Shell
- **Applied AI and Digital Engineering:** Machine Learning, Computer Vision, Workflow Automation, Edge Computing, Data Processing, AI Model Deployment
- **Systems and Controls:** Control Systems, Real-Time Data Systems, Industrial Automation, Systems Monitoring, Data Ingestion, Performance Analysis
- **Professional Competencies:** Analytical Judgment, Technical Problem-Solving, Initiative, Cross-Disciplinary Collaboration, Documentation Quality, Continuous Improvement, Technical Communication

Professional Experience

Hatch Ltd. – *Mechanical Engineer / Package Engineer / Commissioning Engineer / Software Developer*

Toronto, ON | Jan 2020 – Jul 2026

- Contributed to multidisciplinary engineering and project execution activities across mineral processing, lithium and copper processing, energy systems, and industrial infrastructure projects.
- Supported package engineering activities for projects including Lilac, Li-Cycle, and Manjar Maju Refinery, contributing to engineering coordination and the organization of technical project deliverables.
- Contributed to tunnel fire ventilation design for the La Fontayne Tunnel and other projects, gaining experience with mechanical engineering considerations in infrastructure and fire-safety applications.
- Supported commissioning and project execution for the large-scale Jansen potash project as a Handover Coordinator within the HBJV commissioning team.
- Coordinated system handover activities and technical documentation required to transition completed systems into commissioning and operations.
- Prepared Handover Packages, Subsystem Test Packages, and Notices of Energization (NOEs) to support commissioning execution, systems readiness, and project handover.
- Coordinated information and deliverables across engineering, construction, commissioning, and project stakeholders to support project execution and resolve documentation gaps.

- Managed technical project documentation using SmartPlant Foundation, Smart Completions, and SharePoint, supporting information traceability and collaboration across project teams.
- Developed automation for repetitive project documentation activities, including automated generation of Notices of Energization, improving consistency and reducing manual administrative effort.
- Collaborated with multidisciplinary teams to identify technical and workflow issues, improve information management, and support practical engineering solutions.
- Contributed to VISTA, a real-time hydro energy plant operating and optimization system developed using C++ and C#, integrating software functionality with industrial control systems.
- Implemented data-ingestion and performance-analysis modules and maintained legacy software components supporting industrial system operations.
- Modernized legacy C++ components and participated in software development, testing, and deployment activities within an Azure DevOps-driven environment.
- Developed and deployed computer vision capabilities for the Onaping Depth Mine, applying machine learning and industrial data processing to safety and operational-efficiency applications.
- Built and deployed edge computing systems for field-level monitoring and machine-learning inference in industrial environments.
- Applied analytical problem-solving and automation techniques to improve internal engineering and technical workflows.
- Worked across engineering and software development activities to improve technical processes, support system integration, and deliver practical solutions to complex industrial challenges.

SELECTED PROJECTS

Jansen Potash Project — Commissioning and Systems Handover

Hatch / HBJV

- Worked on-site as a Handover Coordinator within the HBJV commissioning team on a large-scale industrial project.
- Coordinated systems handover activities and technical deliverables to support commissioning and operational readiness.
- Prepared Handover Packages, Subsystem Test Packages, and Notices of Energization.
- Coordinated project information across engineering, construction, and commissioning stakeholders.
- Used SmartPlant Foundation, Smart Completions, and SharePoint to manage project documentation and support field execution.
- Automated repetitive documentation workflows, including NOE generation, to improve process consistency and efficiency.
- Supported project execution by organizing deliverables, identifying documentation gaps, and coordinating information required for system handover.

Package Engineering — Mineral Processing Projects

Hatch

- Contributed to package engineering activities for lithium and copper processing projects, including Lilac, Li-Cycle, and Manjar Maju Refinery.
- Supported the coordination and organization of engineering information and technical project deliverables.
- Worked within multidisciplinary engineering environments requiring coordination between technical requirements, project documentation, and execution activities.
- Developed experience contributing to engineering projects involving complex industrial systems and multiple technical stakeholders.

La Fontayne Tunnel — Fire Ventilation Design

Hatch

- Contributed to tunnel fire ventilation design activities.
- Gained exposure to mechanical ventilation engineering considerations associated with infrastructure and fire-safety applications.
- Supported engineering work within a multidisciplinary project environment.
- Developed experience relevant to mechanical system performance and the coordination of technical requirements for infrastructure projects.

VISTA — Energy Plant Optimization Tool

Hatch

- Co-developed a real-time hydro energy plant operating and optimization system using C++ and C#.
- Implemented data-ingestion and performance-analysis functionality for industrial system monitoring.
- Worked with control logic and industrial automation system interfaces.
- Maintained and modernized legacy software components supporting energy plant operations.
- Collaborated within an engineering and software development team using Azure DevOps for development, testing, and deployment.

Onaping Depth Mine — Computer Vision Platform

Hatch

- Developed computer vision capabilities using Python, Flask, PyTorch, and Angular for industrial mining applications.
- Implemented camera-based inspection and analysis functionality using machine learning models.
- Supported deployment of machine learning applications in industrial environments.
- Applied software engineering and analytical techniques to develop practical solutions for operational efficiency and safety-related applications.

Selected Technical and Automation Projects

VISP — Computer Vision Platform on Edge Devices

Personal Project

- Designed and developed an on-premises platform for deploying computer vision applications on edge devices.
- Implemented functionality for managing RTSP cameras and querying stored detections.
- Designed a modular architecture supporting higher-level applications such as alerts, analytics, and operational dashboards.
- Combined machine learning, networking, databases, and web services into an integrated deployment platform.

Custom C++ Framework

Personal Project

- Built a lightweight HTTP server and in-memory database engine in C++.
- Developed practical knowledge of software architecture, networking, data structures, and memory management.
- Applied structured problem-solving to the design and implementation of systems-level software.

Resume Automation

Personal Project

- Developed an automated workflow to convert a Markdown resume into a professionally formatted PDF.
- Applied scripting and document-generation techniques to improve consistency and reduce repetitive formatting work.

Education

University of Toronto — Bachelor of Applied Science in Mechanical Engineering

Minor in Robotics and Mechatronics | Toronto, ON | 2020

- Hatch Engineering Entrance Scholarship — Full Ride Scholarship Recipient.
- Developed foundational knowledge of mechanical engineering, systems integration, robotics, and mechatronics through undergraduate education.

Continuous Education

- **Harvard CS50:** Completed CS50x, CS50 Python, CS50 Web, and CS50 SQL.
- **edX:** Completed online coursework in Data Structures and Algorithms.
- **Boot.dev:** Completed programming coursework involving guided and independent projects.

- **Nand2Tetris:** Completed coursework covering computer architecture, machine language, assembly, compiler development, and operating-system fundamentals.

Interests

Building Engineering, Mechanical Systems Design, Industrial Infrastructure, Fire Ventilation, Sustainable Engineering, Energy Systems, Multidisciplinary Project Delivery, Commissioning, Applied AI in Engineering, Engineering Workflow Automation, Industrial Automation, and Systems Integration.